

Urban Heat and Vegetation Change in Zurich, 1985–2024

Moritz Birle SDS210 Programming with Spatial Data Spring Semester 2026

1. Research Question and Aim

This project investigates the relationship between surface-related raster values (Band 1) and vegetation density (NDVI) in Zurich between 1985 and 2024, using 14 Landsat composite satellite imagery files. The core research question is: How have Band 1 values and NDVI evolved over four decades, and is there a measurable correlation? The analysis further asks whether certain urban districts (Stadtkreise) show stronger vegetation increases than others. Answering this matters because vegetation can reduce surface temperatures through shading and evapotranspiration; documenting the NDVI-Band 1 relationship provides empirical evidence on urban heat mitigation in a growing city.

2. Workflow and Implementation

The analysis was implemented in Python using rasterio, numpy, pandas, matplotlib, geopandas, and osmnx. All 14 raster files (LandsatComposite_Zurich_*.tif) were loaded from Google Drive; each contains 7 spectral bands. Band 1 was used as the primary surface indicator. NDVI was calculated from Band 3 (Red) and Band 4 (NIR) using $NDVI = (NIR - Red) / (NIR + Red)$.

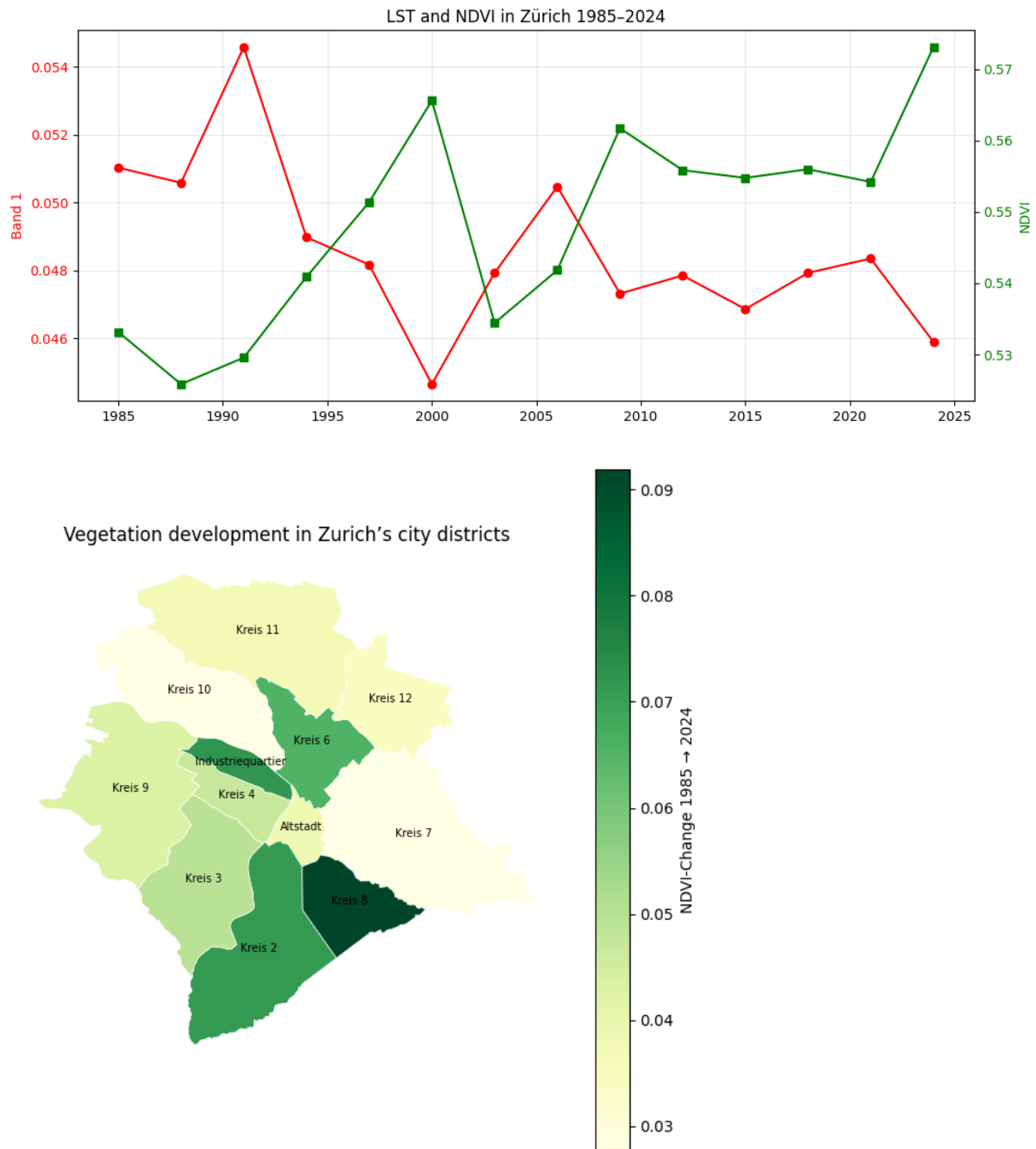
For each year, mean Band 1 and mean NDVI were computed. A three-year rolling mean was applied to highlight the long-term trend. A Pearson correlation was calculated between the two time series. Zurich's 12 Stadtkreise were loaded from OpenStreetMap via osmnx. For each district, zonal means of Band 1 and NDVI were extracted for every year using raster clipping (rasterio.mask.mask). The change between 1985 and 2024 was then mapped.

3. Key Results

Band 1 values declined from approximately 0.030 in 1985 to 0.026 in 2024, with a pronounced drop in the early 1990s and stabilisation since the early 2000s. NDVI increased consistently across all districts. The Pearson correlation between annual mean Band 1 and annual mean NDVI is $r = -0.8221$, indicating a strong negative relationship (see Figure 1 in the appendix).

District-level analysis reveals spatially uneven vegetation growth. Kreis 8 (Seefeld/Weinegg), Kreis 2 (Wollishofen/Enge), and Kreis 6 (Unterstrass/Oberstrass) show the strongest NDVI increases (+0.05 or more), while the Altstadt and Kreis 4 show more moderate gains (+0.03 or less). The results support the assumption that urban greening correlates with lower surface-related values and that greening has been spatially uneven across Zurich.

Figure 1



4. Challenges and Reflection

The main technical challenge was correctly aligning the raster data with the district boundaries. The Landsat composites use a different CRS than the OSM boundary data, so every raster had to be reprojected before clipping. An early version of the loop failed silently for edge districts, returning NaN values. I resolved this by wrapping the clipping step in a try/except block and verifying output shapes for every district-year combination. This taught me to always validate spatial alignment after CRS transformations.

An analytical limitation is the uncertainty about what Band 1 physically represents. The values do not match typical Land Surface Temperature in Kelvin; Band 7 may be the thermal band instead. Band 1 is therefore described as a surface-related raster indicator rather than LST. Atmospheric effects, sensor differences, and cloud contamination across 14 acquisition dates are additional uncontrolled uncertainties.

5. Repository and AI Use

Code and notebook:

<https://github.com/moribir-collab/SDS210-Birle>

AI disclosure: AI tools (ChatGPT) were used during this project for code structuring, documentation assistance, and report drafting. All AI-generated code and text were critically reviewed, edited, and validated. The analytical decisions are my own.